

The Co-Living between Reality and Virtuality as a Daily Routine

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Abstract

Mixed Reality (MR) is emerging as a new standard of reality, where the blending of virtual and physical elements is set to become the norm. With AI-powered XR wearables being more widely available to the public, users will be able to transition from full Virtual Reality (VR) immersion to augmented environments, and perform hybrid interactions with real and virtual elements. Although many participants in this workshop may already have substantial experience with immersive technologies, introducing a broader and more diverse population is still a challenge. An inclusive XR design must take into account different backgrounds and needs, from generational gaps and cultural contexts to variations in abilities and neurodivergence. This workshop invites researchers, designers and practitioners to collectively explore interactive techniques and strategies that support a wide range of contexts and experiences. Through guided discussion, ideas, sparking debates and prototype showcases, our workshop aims to identify strategies for a smoother transition into an MR-centric society.

*Both authors contributed equally to this research and share first authorship.

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CHI EA '26, Barcelona, Spain
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ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3778711>

Keywords

CHI, XR, Reality, Virtual Reality, Immersive UI, Passthrough, Mixed Reality, Accessibility

ACM Reference Format:

Georgios Tsampounaris, Alex Pau Fuentes Raventos, Michael Wiesing, Roger Montserrat Jovellar, Kris Pilcher, Katerina El Raheb, Artur Pilacinski, and Mel Slater. 2026. The Co-Living between Reality and Virtuality as a Daily Routine. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3778711>

1 Motivation

We are reaching a pivotal moment in human history where Mixed Reality (MR) is emerging as a new standard of reality [6, 12]. In this not-so-distant future, virtual and physical elements will blend [17, 22], reshaping how we perceive and interact with our surroundings. Moreover, as AI-Powered Extended Reality (XR) wearables achieve increasing commercial viability [10, 21], interactions in MR are expected to be part of everyday life. This means that actions such as full Virtual Reality (VR) immersion, switching back to reality, isolating virtual objects or subjects in reality, and bringing real objects or subjects into virtuality may become routine [1, 9, 14].

Most conference participants already have substantial exposure to immersive technologies. This is an advantage, as experienced users not only understand the nuances of MR but also know from practice where newcomers often struggle [19]. Such perspectives can therefore help us better identify key challenges in onboarding and interaction design. At the same time, we will emphasize that

these solutions must also address novices with no previous experience with this technology, who may nevertheless encounter MR in everyday contexts such as work, healthcare, education. Along the generational gap between groups of people, there is also a technological gap [13, 18]. The cohorts differ markedly in their prior exposure to interactive technologies. Older adults often report lower levels of digital access and skills, whereas children often acquire touchscreen familiarity in early childhood [4, 5].

In addition to generational differences, inclusivity in this new technological era should also take into account several other contexts, such as culture, disabilities and neurodivergence [2, 3, 20]. Since most Human-Computer Interaction (HCI) research and development is conducted in Western and industrialized contexts [7, 16], many cultural groups are overlooked during the design and evaluation of XR systems [8, 20]. Similarly, users with disabilities face significant barriers such as the lack of captions and auditory descriptions, limited input options for those with reduced mobility, and interfaces that are not adapted to cognitive impairments [2, 3]. Finally, neurodivergent users (e.g. Autism, ADHD) are also often excluded [11], since XR environments can be overwhelming, unpredictable, and inflexible, lacking enough possibilities to adjust sensory input or pacing to individual preferences [15]. Our workshop explicitly builds on this growing body of accessibility and neurodiversity-focused research, and aims to connect it with everyday MR design.

We aim to cover this inclusivity discussion by encouraging content submissions from experts from disciplines such as occupational therapy, physical therapy, special education, psychology, and assistive technology. These challenges form the basis of our workshop, where we aim to design onboarding and interaction methods that accommodate a broad spectrum of digital literacies, needs, and backgrounds. Therefore, we will make a collective effort to think of ways of creating intuitive scenarios and user interfaces to address a smooth and inclusive onboarding of humanity to this transitioning way of living and become more familiar with it. To this end, participants are encouraged to engage with their own subtopics but also express their critical thoughts. As organizers, we will provide a set of questions and prototypes to spark discussion and creativity. Prototypes from participants are more than welcome to be presented and discussed, regardless of whether they can be applied in real life or not.

2 Length of Workshop

Long workshop (two sessions) - 180 minutes, 90 minutes each session.

3 Organizers

Georgios Tsampounaris is an XR developer and visual artist. His work spans XR development (AR/VR), audio-reactive and real-time rendered visuals, ensuring projects unite technical innovation with artistic depth. Georgios is active in both academia and the creative industry: publications at SIGGRAPH, utilizing immersive VFX for haptic feedback and teaching technical workflows for VR cinematography in academic contexts. In the creative sector, he has contributed internationally as a past member of Ars Electronica's Futurelab, collaborating on cutting-edge media art and immersive experiences in

Deepspace 8K. His expertise includes creative direction, 3D scanning, motion capture workflows, and non-proprietary AI-driven generative visuals. This unique cross-disciplinary background allows him to connect research, education, and cultural storytelling with innovative XR and audiovisual practices. He is founder 70% Water immersive studio.

Alex Fuentes Raventós is a PhD student at the Autonomous University of Barcelona and a Research Technician at the Event Lab, University of Barcelona, where he is also a member of the Institute of Neurosciences. He is one of the Coordinators of the Deep Tech Brain Program at the University of Barcelona. His research interests focus on how Virtual and Mixed Reality can influence people's behaviors. His work lies at the intersection of Artificial Intelligence and Neuroscience, always using the VR headset as his medium.

Michael Wiesing is a postdoctoral researcher at the Event Lab (University of Barcelona) whose work bridges cognitive neuroscience and immersive technologies. His research focuses on measuring presence and co-presence in VR. He develops VR experiments and technical tooling in Unity, integrating haptics, timing/trigger pipelines, and real-time rendering. He co-organizes the annual VR Summit and its Hackathon, publishes CC0 virtual environments to support open science, and collaborates within the EU PRESENCE project on presence evaluation. His portfolio includes empirical studies, open-source releases, and technical workshops that connect research, education, and applied XR development.

Art Pilacinski has background in human neural and behavioral sciences specializing in human motor control and sensorimotor cognition. He received his PhD from University of Tuebingen, Germany. He now works at the Department for Neurotechnology at Ruhr University Bochum, Germany. There, his research is focused on phantom touch in virtual reality, number processing in single neurons of human intraparietal sulcus, non-visual eye movements during thinking, intuitive biomechanics and collaborative robotics. Last but not least, he runs the VR Summit conference.

Roger Montserrat Jovellar is a General Health Psychologist and PhD student at the Event Lab, University of Barcelona. He began his research career investigating ageing and psychosis across different institutions, before eventually discovering the potential of VR for psychology. Since then, his work has focused on VR in areas such as exposure and reminiscence therapy, later incorporating AI into his work as LLMs became viable. His doctoral thesis brings together this trajectory by using VR and AI to train clinical psychologists.

Kris Pilcher is a creative technologist and researcher exploring AI, performance, and sensory design. Working between Cambridge and Barcelona, he combines experimental AI systems with embodied interfaces to investigate how machines can dream, feel, and evolve new ontologies. He is a Research Affiliate at MIT's Spatial Sound Lab and Artist-in-Residence at Georgia State University's Creative Media Industries Institute. His projects—including PIP: Purposefully Induced Psychosis, AI JOE, and Aletheia—have been presented at CHI, Ars Electronica, VRHAM, CCCB Barcelona, and Cannes XR. A winner of MIT StageHack, three-time MIT Reality Hack awardee, and Oculus Start/Launchpad developer, he lectures internationally on XR, AI art, and posthuman design, with collaborations ranging from Berkeley SETI to Roscosmos partners.

Katerina El Raheb is an Assistant Professor at the Department of Performing and Digital Arts, University of Peloponnese, specializing in Informatics Applications in Performing Arts. Since 2009, she has collaborated as an HCI researcher with the MADgIK Lab (NKUA) and Athena Research Center, working on EU and national projects in cultural heritage, artistic creation, and education (e.g., BRIDGES, Aria, Transition to 8, DERC, WhoLoDancE). She holds a PhD in Informatics (NKUA) on dance representation and interaction, an MSc in Advanced Information Systems, and a BSc in Chemical Engineering (NTUA). A trained dancer with a diploma from Niki Kontaxaki dance school, she has performed with groups such as Yelp Dance Co., Periigites, Kinitiras, and as an EMAP resident artist at Onassis Center.

Mel Slater, DSc, is a Distinguished Investigator at the University of Barcelona in the Institute of Neurosciences, and co-Director of the Event Lab (Experimental Virtual Environments for Neuroscience and Technology). He was previously Professor of Virtual Environments at University College London in the Department of Computer Science. He has been involved in research in virtual reality since the early 1990s and his work has concentrated on both technical developments in VR and contributions to the understanding of presence and the cognitive neuroscience of body ownership and agency. He has been first supervisor of 41 PhDs in graphics and virtual reality. He was a Research Award Winner of the Alexander von Humboldt Foundation in 2021 and was elected to the IEEE VGTC Virtual Reality Academy in 2022. He is the Field Editor of *Frontiers in Virtual Reality*, and Chief Editor of the *Human Behaviour in Virtual Reality* section. He has had the goal of bringing virtual reality research to an interdisciplinary audience, with publications in journals such as *Nature Reviews Neuroscience*, *Neuron*, *Cortex*, *The Lancet Psychiatry*, as well as in specialist journals such as *ACM Transactions on Graphics*, and *IEEE Transactions on Visualization and Computer Graphics*. He is a co-founder of Kiin.tech.

4 Plans to Publish Workshop Proceedings

To ensure that contributions and discussions extend beyond the day of the event, we will adopt a multi-pronged dissemination strategy:

- **Proceedings:** Accepted papers, posters, and demo abstracts will be collected and published in CEUR-WS, ensuring that they are permanently accessible and citable.
- **Workshop Website:** All accepted submissions, together with presentation slides, images of prototypes, and a summary of group discussions, will be hosted on the workshop website. This will serve as a living archive and a point of reference for future collaborations.
- **Open repository mirror:** All accepted submissions, slides, and supporting materials will also be archived on Zenodo, assigning DOIs for long-term citation and reuse.
- **Workshop Report:** The organizers will prepare a post-workshop report highlighting key outcomes, insights, and design principles emerging from the event. This report will be submitted to venues such as the ACM Interactions blog or SIGCHI Medium to reach the wider HCI community.

- **Optional Journal Publication:** Depending on the level of interest and quality of submissions, we will explore the possibility of inviting extended versions of selected contributions to a special issue in a relevant journal.

5 Accessibility

The workshop will take place on-site at CHI 2026 in Barcelona. We are committed to making the event accessible and inclusive for all participants. Our accessibility plans include:

- **Physical access:** We will coordinate with the CHI organizers to ensure that the workshop venue is wheelchair accessible and provides flexible seating arrangements. Group activities will be designed so participants can contribute without needing to stand or move excessively.
- **Accessible materials:** Accepted papers, posters, and slides will be made available in advance on the workshop website as accessible PDFs with embedded text. Posters will be accompanied by short text summaries, and demo descriptions will include captions or alt-text for images.
- **Sensory accessibility:** Instructions during activities will be given in both spoken and written form. We will avoid relying on audio-only or visual-only cues.
- **Inclusive facilitation:** Group activities will allow for varied forms of participation, such as note-taking, sketching, or presenting, so that participants with different abilities can contribute effectively. Organizers will actively monitor participation to ensure all voices are included.
- **Support requests:** Participants will be invited to share any specific accessibility needs in advance, and we will adapt facilitation and materials wherever possible.

By focusing on physical access, accessible formats, sensory considerations, and inclusive facilitation, we aim to ensure that the workshop is welcoming and usable for participants with diverse abilities and backgrounds.

6 Offline Materials

To ensure that the outcomes of the workshop are accessible beyond the conference day, we will make all materials openly available through the workshop website. This will include:

- **Proceedings and submissions:** Accepted papers, posters, and demo abstracts will be shared as accessible PDFs with embedded text.
- **Presentation slides and prototypes:** Organizers and participants will be invited to upload their slides, images, and demo descriptions. Where possible, short video recordings or animated walkthroughs of prototypes will also be provided.
- **Workshop summary:** A collective summary highlighting key debates, co-design outcomes, and design principles will be published after the event. This will complement the official workshop report described in the Publishing Plans section.
- **Discussion archive:** Notes and visual materials from breakout groups (e.g., sketches, Miro boards, or photos of sticky notes) will be digitized and shared online, preserving both ideas and process.

All materials will be hosted on the workshop website (<https://colivingvr.github.io/CHI2026/>) and mirrored on Zenodo to ensure long-term accessibility and citation. This will allow participants, as well as the broader CHI community, to revisit the discussions, re-use the materials in teaching and research, and build on the outcomes of the workshop.

7 Workshop Activities

The workshop will be highly interactive, combining short framing insights from the organizers, demonstrations of prototypes, presentations of accepted contributions, and group discussions. We aim to have a long workshop of 180 minutes.

The goal is to move from critical reflection to creative prototyping of everyday MR experiences.

1. Opening and Scene Setting (15 min). We begin with an introduction by the organizers, framing XR as a technology that is rapidly moving toward everyday use, similar to the adoption of smartphones. Two guiding questions will shape the day: (1) How can transitions between reality and virtuality become seamless and intuitive? (2) How do we ensure inclusiveness for groups less familiar with new technologies?

2. Paper Presentations (50 min). Authors of accepted short papers and position papers will briefly present their work (5–7 minutes each). These contributions will help seed the discussion with multiple perspectives, ranging from technical prototypes to conceptual reflections.

3. Debate and Critical Reflections (25 min). A moderated debate will invite participants to discuss provocative questions, such as: – Do we really want to onboard everyone into XR, or should disengagement remain an option? – What are the ethical implications of XR wearables continuously recording? This debate ensures that ethical, social, and accessibility concerns remain central to the workshop.

– **Break (15 min)** –

4. Prototype Demonstrations and Posters (20 min). Accepted demos and posters will be presented in an interactive format. For example, prototypes may include scenarios such as a user carrying a virtual object across reality/virtuality transitions, or scalable text overlays for accessibility. These demonstrations will serve as tangible entry points for further group work.

5. Breakout Groups and Co-Design (40 min). Participants will form small groups to sketch or prototype their own XR interactions or design concepts. Contributions can be inspired by accepted submissions, personal projects, or speculative ideas. The emphasis will be on inclusiveness, everyday usability, and ethics. Organizers will provide seed questions and materials to support brainstorming.

6. Sharing and Synthesis (20 min). The groups will present their concepts to the full workshop. Together, we will extract common design principles and identify open research challenges for seamless, ethical, and inclusive MR.

7. Wrap-up and Next Steps (10 min). The workshop will conclude with a reflection on the insights and outcomes. We will outline plans for publishing a workshop report, summarizing the design principles, debates, and contributions. Participants will also be invited to continue collaborating via the workshop website and mailing list.

8 Call for Participation (250 words)

This workshop explores how Mixed Reality (MR) is becoming embedded in everyday routines and asks how we can design seamless, inclusive, and meaningful transitions between the physical and the virtual. XR wearables are moving toward daily use, much like smartphones a decade ago. How can we ensure that these interactions feel natural, intuitive, and accessible for all, including first-time adopters and underrepresented groups?

We invite contributions that address the following themes:

- **Seamless Transitions:** Design metaphors, interface concepts, and technical solutions that allow users to intuitively move between VR MR, and the physical world.
- **Onboarding and Accessibility:** Approaches to make XR inclusive for groups with all types of profiles, independently of culture, disabilities, or limited technological backgrounds.
- **Embodied and Everyday Interaction:** User interfaces, techniques, and prototypes that integrate XR into daily activities such as cooking, commuting, or healthcare.
- **Speculative and Critical Perspectives:** Reflections on opting in -or opting out- of XR in societies where it becomes the norm.
- **Prototypes and Demonstrations:** Systems, mock-ups, or artworks that explore future scenarios and metaphors.

We welcome **short papers (up to 6 pages)**, **position papers (2–4 pages)**, and **poster/demo abstracts (1 page)**. Submissions will undergo a lightweight review to ensure quality and relevance. Accepted contributions will be published in CEUR-WS, mirrored on Zenodo, and posted on the workshop website. At least one author of each accepted submission must register and attend the workshop.

Join us to imagine futures where the boundaries between worlds dissolve -yet users remain comfortable, guided, and empowered.

9 Expected size of attendance

We expect 15–25 participants, ensuring an intimate but diverse community.

Acknowledgments

This workshop is supported by the following projects:

Presence XR, a Horizon Europe Innovation Project co-financed by the EC under Grant Agreement ID: 101135025

environMENTAL, Funded by the European Union (Grant agreement No 101057429). Complementary funding was received by UK Research and Innovation (UKRI) under the UK government’s Horizon Europe funding guarantee (10131373 and 10038599) and the National Key R&D Program of the Ministry of Science and Technology of China (MOST 2023YFE0199700).

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